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Inventor(s)

Hasan; Mirmd Rashedul et al.

SYSTEMS FOR DESCRIBING UNKNOWN ACCESS MANAGEMENT EVENTS USING IDENTITY TAGS AND RELATED TRANSACTION CHAINS

Abstract

The technology includes a system to describe access management events. The system monitors transactions performed by layers of an Identity Provider service (IDP). Each layer corresponds to an Application Programming Interface (API). The system captures output from the layers in a dynamic record. The output includes timestamps, names of layers, and principals responsible for calling layers. The system determines related transactions by tagging output with identity tags based on principals, and grouping transactions based on common identity tags. The system chains related transactions into lists by comparing identity tags and timestamps and determining orders based on proximate timestamps. The system determines access management events based on names of layers from lists of chained transactions. The system generates descriptions of these events for display to auditors by using language models which have been provided input from the dynamic record.

Inventors: Hasan; Mirmd Rashedul (Renton, WA), Medeiros Dominguez; Pablo (Redmond, WA), Srivastava; Nilay (Sammamish, WA)

Applicant: T-Mobile USA, Inc. (Bellevue, WA)

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Background/Summary

BACKGROUND

[0001] Identity and access management (IAM) is a framework of policies and technologies to ensure that the right users have the appropriate access to technology resources. IAM systems identify, authenticate, and control access for individuals who will be utilizing IT resources, as well as hardware and applications needed for access. IAM covers issues such as how users gain an identity, the roles and, sometimes, the permissions that identity grants, the protection of that identity, and the technologies supporting that protection (e.g., network protocols, digital certificates, passwords, etc.). IAM systems, products, applications, and platforms manage identifying and ancillary data about entities that include individuals, computer-related hardware, and software applications. This includes identity providers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Detailed descriptions of implementations of the present invention will be described and explained through the use of the accompanying drawings.

[0003] FIG. 1 is a block diagram that illustrates a wireless communications system that can implement aspects of the present technology.

[0004] FIG. 2 is a block diagram that illustrates 5G core network functions (NFs) that can implement aspects of the present technology.

[0005] FIG. 3 is a block diagram that illustrates a system for contextualizing access management events with identity tags and related transaction chains.

[0006] FIG. 4A is a block diagram that illustrates components of a system for components of a system for capturing, tagging, and chaining output from transactions performed by API layers of an IDP.

[0007] FIG. 4B is a block diagram that illustrates components of a system for describing an unknown access management event using identity tags and related transaction chains.

[0008] FIG. 5 is a flowchart that illustrates a method to contextualize access management events with identity tags and related transaction chains.

[0009] FIG. 6 is a block diagram that illustrates an example of a computer system in which at least some operations described herein can be implemented.

[0010] The technologies described herein will become more apparent to those skilled in the art from studying the Detailed Description in conjunction with the drawings. Embodiments or implementations describing aspects of the invention are illustrated by way of example, and the same references can indicate similar elements. While the drawings depict various implementations for the purpose of illustration, those skilled in the art will recognize that alternative implementations can be employed without departing from the principles of the present technologies. Accordingly, while specific implementations are shown in the drawings, the technology is amenable to various modifications.

DETAILED DESCRIPTION

[0011] The disclosed technology relates to describing unknown access management events for an identity provider (IDP). In some implementations, the IDP is a service that controls access to a system and its functions by performing access management events—these events include

identifying, authenticating, and controlling a user's degree of access to a system, as well as functions performed by other services comprised by the system. In this application, access management events are also referred to as access events, or, simply, "events." A record of these events is kept for the purpose of preventing fraud, auditing user identities, and building new data-driven services and tools for users of the system. Such services and tools can include Machine Learning (ML) and Artificial Intelligence (AI) applications.

[0012] In some implementations, an access event relies on subsidiary Application Programming Interfaces (API) to perform services that are interconnected or interdependent. These subsidiary API services can be referred to as "transactions." For example, in order for a user to add a new phone line to their profile, a first transaction identifies which user is making the request, a second transaction verifies that the user has the proper authority to be granted that request, a third transaction grants the request, and a fourth transaction notifies the user of the granted request. In such examples, the output from one transaction can serve as the input for one or more other subsidiary APIs. It should be understood that an access management can include more or fewer subsidiary API services, or transactions.

[0013] The disclosed technology can provide a solution to the problem presented by current IDP logs, in which the subsidiary API services (i.e., transactions) are either not recorded, or recorded separately. As a result, current IDP logs contain records that are cryptic and unusable to all but a select few experts, and which require a great deal of time and expense to combine the data repositories where transactions are recorded separately. These data repositories are multivarious, comprising their own specific data types, security protocols, APIs, locations, modes of storage, and means of extraction, all of which must be reconciled before the data which they comprise can be combined. The disclosed technology can solve these problems, and others, by continually updating a consolidated record of all transactions performed under an access management event. The disclosed technology can accomplish this by chaining related transactions, and grouping chained transactions according to identity tags. For example, identity tags are related to an originating principal. In an IDP, entities are referred to as principals. Principals can include a person, a user, a subscriber, a computer, a service, or a computational entity, such as a process, or a thread. From the consolidated record of chained transactions with identity tags, the system can create a single comprehensive view of the IDP, arranged chronologically. The single comprehensive view of the IDP can function as a basis for AI and ML applications, for the purpose of continual anomaly detection among access management events, as well as automatic intent prediction, and service recommendation, among others.

[0014] The description contained herein, and the associated drawings, are illustrative examples and are not to be construed as limiting. This disclosure provides certain details for a thorough understanding and enabling description of these examples. One skilled in the relevant technology will understand, however, that the invention can be practiced without many of these details. Likewise, one skilled in the relevant technology will understand that the invention can include well-known structures or features that are not shown or described in detail, to avoid unnecessarily obscuring the descriptions of examples.

Wireless Communications System

[0015] FIG. 1 is a block diagram that illustrates a wireless telecommunication network **100** ("network **100**") in which aspects of the disclosed technology are incorporated. The network **100** includes base stations **102-1** through **102-4** (also referred to individually as "base station **102**" or collectively as "base stations **102**"). A base station is a type of network access node (NAN) that can also be referred to as a cell site, a base transceiver station, or a radio base station. The network **100** can include any combination of NANs including an access point, radio transceiver, gNodeB (gNB), NodeB, eNodeB (eNB), Home NodeB or Home eNodeB, or the like. In addition to being a wireless wide area network (WWAN) base station, a NAN can be a wireless local area network (WLAN) access point, such as an Institute of Electrical and Electronics Engineers (IEEE) 802.11 access

point.

[0016] The NANs of a network **100** formed by the network **100** also include wireless devices **104-1** through **104-7** (referred to individually as “wireless device **104**” or collectively as “wireless devices **104**”) and a core network **106**. The wireless devices **104** can correspond to or include network **100** entities capable of communication using various connectivity standards. For example, a 5G communication channel can use millimeter wave (mmW) access frequencies of 28 GHz or more. In some implementations, the wireless device **104** can operatively couple to a base station **102** over a long-term evolution/long-term evolution-advanced (LTE/LTE-A) communication channel, which is referred to as a 4G communication channel.

[0017] The core network **106** provides, manages, and controls security services, user authentication, access authorization, tracking, internet protocol (IP) connectivity, and other access, routing, or mobility functions. The base stations **102** interface with the core network **106** through a first set of backhaul links (e.g., S1 interfaces) and can perform radio configuration and scheduling for communication with the wireless devices **104** or can operate under the control of a base station controller (not shown). In some examples, the base stations **102** can communicate with each other, either directly or indirectly (e.g., through the core network **106**), over a second set of backhaul links **110-1** through **110-3** (e.g., X1 interfaces), which can be wired or wireless communication links.

[0018] The base stations **102** can wirelessly communicate with the wireless devices **104** via one or more base station antennas. The cell sites can provide communication coverage for geographic coverage areas **112-1** through **112-4** (also referred to individually as “coverage area **112**” or collectively as “coverage areas **112**”). The coverage area **112** for a base station **102** can be divided into sectors making up only a portion of the coverage area (not shown). The network **100** can include base stations of different types (e.g., macro and/or small cell base stations). In some implementations, there can be overlapping coverage areas **112** for different service environments (e.g., Internet of Things (IoT), mobile broadband (MBB), vehicle-to-everything (V2X), machine-to-machine (M2M), machine-to-everything (M2X), ultra-reliable low-latency communication (URLLC), machine-type communication (MTC), etc.).

[0019] The network **100** can include a 5G network **100** and/or an LTE/LTE-A or other network. In an LTE/LTE-A network, the term “eNBs” is used to describe the base stations **102**, and in 5G new radio (NR) networks, the term “gNBs” is used to describe the base stations **102** that can include mmW communications. The network **100** can thus form a heterogeneous network **100** in which different types of base stations provide coverage for various geographic regions. For example, each base station **102** can provide communication coverage for a macro cell, a small cell, and/or other types of cells. As used herein, the term “cell” can relate to a base station, a carrier or component carrier associated with the base station, or a coverage area (e.g., sector) of a carrier or base station, depending on context.

[0020] A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and can allow access by wireless devices that have service subscriptions with a wireless network **100** service provider. As indicated earlier, a small cell is a lower-powered base station, as compared to a macro cell, and can operate in the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Examples of small cells include pico cells, femto cells, and micro cells. In general, a pico cell can cover a relatively smaller geographic area and can allow unrestricted access by wireless devices that have service subscriptions with the network **100** provider. A femto cell covers a relatively smaller geographic area (e.g., a home) and can provide restricted access by wireless devices having an association with the femto unit (e.g., wireless devices in a closed subscriber group (CSG), wireless devices for users in the home). A base station can support one or multiple (e.g., two, three, four, and the like) cells (e.g., component carriers). All fixed transceivers noted herein that can provide access to the network **100** are NANs, including small cells.

[0021] The communication networks that accommodate various disclosed examples can be packet-

based networks that operate according to a layered protocol stack. In the user plane, communications at the bearer or Packet Data Convergence Protocol (PDCP) layer can be IP-based. A Radio Link Control (RLC) layer then performs packet segmentation and reassembly to communicate over logical channels. A Medium Access Control (MAC) layer can perform priority handling and multiplexing of logical channels into transport channels. The MAC layer can also use Hybrid ARQ (HARQ) to provide retransmission at the MAC layer, to improve link efficiency. In the control plane, the Radio Resource Control (RRC) protocol layer provides establishment, configuration, and maintenance of an RRC connection between a wireless device **104** and the base stations **102** or core network **106** supporting radio bearers for the user plane data. At the Physical (PHY) layer, the transport channels are mapped to physical channels.

[0022] Wireless devices can be integrated with or embedded in other devices. As illustrated, the wireless devices **104** are distributed throughout the network **100**, where each wireless device **104** can be stationary or mobile. For example, wireless devices can include handheld mobile devices **104-1** and **104-2** (e.g., smartphones, portable hotspots, tablets, etc.); laptops **104-3**; wearables **104-4**; drones **104-5**; vehicles with wireless connectivity **104-6**; head-mounted displays with wireless augmented reality/virtual reality (AR/VR) connectivity **104-7**; portable gaming consoles; wireless routers, gateways, modems, and other fixed-wireless access devices; wirelessly connected sensors that provide data to a remote server over a network; IoT devices such as wirelessly connected smart home appliances; etc.

[0023] A wireless device (e.g., wireless devices **104**) can be referred to as a user equipment (UE), a customer premises equipment (CPE), a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a handheld mobile device, a remote device, a mobile subscriber station, a terminal equipment, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a mobile client, a client, or the like.

[0024] A wireless device can communicate with various types of base stations and network **100** equipment at the edge of a network **100** including macro eNBs/gNBs, small cell eNBs/gNBs, relay base stations, and the like. A wireless device can also communicate with other wireless devices either within or outside the same coverage area of a base station via device-to-device (D2D) communications.

[0025] The communication links **114-1** through **114-9** (also referred to individually as “communication link **114**” or collectively as “communication links **114**”) shown in network **100** include uplink (UL) transmissions from a wireless device **104** to a base station **102** and/or downlink (DL) transmissions from a base station **102** to a wireless device **104**. The downlink transmissions can also be called forward link transmissions while the uplink transmissions can also be called reverse link transmissions. Each communication link **114** includes one or more carriers, where each carrier can be a signal composed of multiple sub-carriers (e.g., waveform signals of different frequencies) modulated according to the various radio technologies. Each modulated signal can be sent on a different sub-carrier and carry control information (e.g., reference signals, control channels), overhead information, user data, etc. The communication links **114** can transmit bidirectional communications using frequency division duplex (FDD) (e.g., using paired spectrum resources) or time division duplex (TDD) operation (e.g., using unpaired spectrum resources). In some implementations, the communication links **114** include LTE and/or mmW communication links.

[0026] In some implementations of the network **100**, the base stations **102** and/or the wireless devices **104** include multiple antennas for employing antenna diversity schemes to improve communication quality and reliability between base stations **102** and wireless devices **104**. Additionally or alternatively, the base stations **102** and/or the wireless devices **104** can employ multiple-input, multiple-output (MIMO) techniques that can take advantage of multi-path environments to transmit multiple spatial layers carrying the same or different coded data.

[0027] In some examples, the network **100** implements 6G technologies including increased

densification or diversification of network nodes. The network **100** can enable terrestrial and non-terrestrial transmissions. In this context, a Non-Terrestrial Network (NTN) is enabled by one or more satellites, such as satellites **116-1** and **116-2**, to deliver services anywhere and anytime and provide coverage in areas that are unreachable by any conventional Terrestrial Network (TN). A 6G implementation of the network **100** can support terahertz (THz) communications. This can support wireless applications that demand ultrahigh quality of service (QOS) requirements and multi-terabits-per-second data transmission in the era of 6G and beyond, such as terabit-per-second backhaul systems, ultra-high-definition content streaming among mobile devices, AR/VR, and wireless high-bandwidth secure communications. In another example of 6G, the network **100** can implement a converged Radio Access Network (RAN) and Core architecture to achieve Control and User Plane Separation (CUPS) and achieve extremely low user plane latency. In yet another example of 6G, the network **100** can implement a converged Wi-Fi and Core architecture to increase and improve indoor coverage.

5G Core Network Functions

[0028] FIG. **2** is a block diagram that illustrates an architecture **200** including 5G core network functions (NFs) that can implement aspects of the present technology. A wireless device **202** can access the 5G network through a NAN (e.g., gNB) of a RAN **204**. The NFs include an Authentication Server Function (AUSF) **206**, a Unified Data Management (UDM) **208**, an Access and Mobility management Function (AMF) **210**, a Policy Control Function (PCF) **212**, a Session Management Function (SMF) **214**, a User Plane Function (UPF) **216**, and a Charging Function (CHF) **218**.

[0029] The interfaces N1 through N15 define communications and/or protocols between each NF as described in relevant standards. The UPF **216** is part of the user plane and the AMF **210**, SMF **214**, PCF **212**, AUSF **206**, and UDM **208** are part of the control plane. One or more UPFs can connect with one or more data networks (DNs) **220**. The UPF **216** can be deployed separately from control plane functions. The NFs of the control plane are modularized such that they can be scaled independently. As shown, each NF service exposes its functionality in a Service Based Architecture (SBA) through a Service Based Interface (SBI) **221** that uses HTTP/2. The SBA can include a Network Exposure Function (NEF) **222**, an NF Repository Function (NRF) **224**, a Network Slice Selection Function (NSSF) **226**, and other functions such as a Service Communication Proxy (SCP).

[0030] The SBA can provide a complete service mesh with service discovery, load balancing, encryption, authentication, and authorization for interservice communications. The SBA employs a centralized discovery framework that leverages the NRF **224**, which maintains a record of available NF instances and supported services. The NRF **224** allows other NF instances to subscribe and be notified of registrations from NF instances of a given type. The NRF **224** supports service discovery by receipt of discovery requests from NF instances and, in response, details which NF instances support specific services.

[0031] The NSSF **226** enables network slicing, which is a capability of 5G to bring a high degree of deployment flexibility and efficient resource utilization when deploying diverse network services and applications. A logical end-to-end (E2E) network slice has pre-determined capabilities, traffic characteristics, and service-level agreements and includes the virtualized resources required to service the needs of a Mobile Virtual Network Operator (MVNO) or group of subscribers, including a dedicated UPF, SMF, and PCF. The wireless device **202** is associated with one or more network slices, which all use the same AMF. A Single Network Slice Selection Assistance Information (S-NSSAI) function operates to identify a network slice. Slice selection is triggered by the AMF, which receives a wireless device registration request. In response, the AMF retrieves permitted network slices from the UDM **208** and then requests an appropriate network slice of the NSSF **226**.

[0032] The UDM **208** introduces a User Data Convergence (UDC) that separates a User Data

Repository (UDR) for storing and managing subscriber information. As such, the UDM **208** can employ the UDC under 3GPP TS 22.101 to support a layered architecture that separates user data from application logic. The UDM **208** can include a stateful message store to hold information in local memory or can be stateless and store information externally in a database of the UDR. The stored data can include profile data for subscribers and/or other data that can be used for authentication purposes. Given a large number of wireless devices that can connect to a 5G network, the UDM **208** can contain voluminous amounts of data that is accessed for authentication. Thus, the UDM **208** is analogous to a Home Subscriber Server (HSS) and can provide authentication credentials while being employed by the AMF **210** and SMF **214** to retrieve subscriber data and context.

[0033] The PCF **212** can connect with one or more Application Functions (AFs) **228**. The PCF **212** supports a unified policy framework within the 5G infrastructure for governing network behavior. The PCF **212** accesses the subscription information required to make policy decisions from the UDM **208** and then provides the appropriate policy rules to the control plane functions so that they can enforce them. The SCP (not shown) provides a highly distributed multi-access edge compute cloud environment and a single point of entry for a cluster of NFs once they have been successfully discovered by the NRF **224**. This allows the SCP to become the delegated discovery point in a datacenter, offloading the NRF **224** from distributed service meshes that make up a network operator's infrastructure. Together with the NRF **224**, the SCP forms the hierarchical 5G service mesh.

[0034] The AMF **210** receives requests and handles connection and mobility management while forwarding session management requirements over the N11 interface to the SMF **214**. The AMF **210** determines that the SMF **214** is best suited to handle the connection request by querying the NRF **224**. That interface and the N11 interface between the AMF **210** and the SMF **214** assigned by the NRF **224** use the SBI **221**. During session establishment or modification, the SMF **214** also interacts with the PCF **212** over the N7 interface and the subscriber profile information stored within the UDM **208**. Employing the SBI **221**, the PCF **212** provides the foundation of the policy framework that, along with the more typical QoS and charging rules, includes network slice selection, which is regulated by the NSSF **226**.

Describing Unknown Access Management Events Using Identity Tags and Related Transaction Chains

[0035] FIG. 3 is a block diagram that illustrates a system **300** for contextualizing access management events **360** with identity tags and related transaction chains. In some implementations, the system **300** can accomplish this by monitoring transactions **332** performed by layers **328** of an Identity Provider service (IDP) **324** for an unknown access management event **360**. Each layer **328** can correspond to a separate Application Programming Interface (API) with its own distinct functionality in the IDP **324**.

[0036] For example, the transactions **332** can include identification, authentication, adjusting a degree of access to the system **300**, and creating a record of the result and of a user associated with the unknown access management event **360**. In some implementations, the system **300** belongs to a network **320**. For example, the network **320** can be a telecommunications network **320**. The user can access the system **300** via an electronic device **304**. The user can perform the unknown access management event **360** using the electronic device **304**, which can be communicated to the network **320** using one or more network access nodes **308** and **312**. In some implementations, the electronic device **304** communicates with a satellite **316** in order to perform the unknown access management event **360** (e.g., verifying user location).

[0037] FIG. 4A is a block diagram that illustrates components of a system **400** for capturing, tagging, and chaining output from transactions **432** performed by API layers **428** of an IDP **424**. One purpose of these operations is to describe access management events that include a common purpose for a grouping of chained related transactions, as well as a result and an identity associated

with a user who either performs the access management event, or on whose behalf a system expert performs the access management event. In some implementations, the layers **428** can include built-in hooks **429**. The built-in hooks **429** can be configured to stream output from transactions **432**. [0038] In some implementations, the system **400** captures output from the layers **428** in a dynamic record **436** of transactions **432**. For example, the output for each transaction **432** includes a timestamp **440**, a name **444** of a layer **428** that performed the transaction, and a principal **448** responsible for calling the layer **428**. The output from the layers **428** can be streamed to the dynamic record **436** of transactions **432** using the built-in hooks **429**.

[0039] In some implementations, the system **400** determines related transactions **432** in the dynamic record **436** by tagging the output for each transaction **432** with identity tags **452** based on the principals **448**. For example, the transactions **432** can be grouped based on common identity tags **452**. Identity tags can include usernames, IDs, or other unique information associated with a single entity or subscriber to the telecommunications network. The transactions **432** can include user identification, authentication, and granting access to the user to select functions of the 5G Core Network illustrated in FIG. 2.

[0040] In some implementations, the principals **448** include source information. The source information can include a user ID, a session ID, a relational ID, a device type, a device ID, an IP address, or a client ID, or any combination or subset of the foregoing. In some implementations, tagging the output for each transaction with identity tags based on the principals **448** also includes parsing the principals for source information. For example, parsing the principals **448** for source information includes determining user profiles that match the source information. This can include searching for the source information in a user profile index or an identity system. In some implementations, the user profiles include identifying information. The system **400** can determine identity tags **452** from the identifying information.

[0041] In some implementations, the system **400** chains the related transactions into a list of chained transactions **433** by comparing identity tags **452** and timestamps **440**. For example, the system **400** can determine an order for the list of chained transactions **433** based on proximate timestamps **440** (e.g., the system **400** can order the list of chained transactions **433** from first to last).

[0042] FIG. 4B is a block diagram that illustrates components of a system **400** for describing an unknown access management event **460** using identity tags and related transaction chains. In some implementations, the system **400** describes an unknown access management event **460** based on the names **444** of the layers **428** of the transactions **432** of the list of chained transactions **433**.

[0043] In some implementations, the names of the layers that performed the transactions are an intention input. For example, determining the access management event **460** can include providing the names of the layers that performed the transactions as the intention input to an intention prediction model. The intention prediction model can be trained to output predicted intentions based on names of layers provided as intention inputs. In some implementations, the system **400** determines a predicted intention from the intention input. For example, the predicted intention can include a common purpose for the transactions **432**, and the system **400** can determine the access management event **460** based on the common purpose and the identity tags **452**. The intention prediction model can be trained by system **400** domain experts based on historical logs of transactions. Such historical logs can include the names of transactions arranged into ordered chains, such as the chained lists of related transactions. Each ordered chain of transactions can be associated with an access management event defined by the domain experts.

[0044] In some implementations, the system **400** generates a description **437** of the access management event **460**, using a model **456** trained to output descriptions **437** based on chained transactions **433** provided as input. The model **456** can be a language model, a large language model (LLM), a deep learning model, or a neural network. The model **456** can include convolutional layers, hidden layers, Long Short Term Memory layers, Transformers, or other

attention mechanisms, or any combination of the foregoing.

[0045] In some implementations, determining the description of the access management event **460** includes organizing the list of chained transactions **433** into a hierarchically nested set of feature-value pairs **435**. The hierarchically nested set of feature-value pairs **435** can include features that are descriptive of the access management event **460**, and values that correspond to the features for each transaction **432**. For example, the hierarchically nested set of feature-value pairs **435** can act as a description input for the model **456**.

[0046] In some implementations, the system **400** generates the description **437** for display to an auditor in a single comprehensive view **468** of the IDP **424**. The single comprehensive view **468** can be updated as new output is streamed to the dynamic record **436** of transactions **432**. The description **437** can provide context for the access management event **460** by identifying a user **453** based on the identity tags **452** and a result **461** based on the access management event **460**.

[0047] In some implementations, the single comprehensive view **468** includes a historical record of previous descriptions **439** of other access management events performed by the user **453**. For example, generating the description **437** for display can include comparing the values of the hierarchically nested set of feature-value pairs **435** against past values of the same features taken from the previous descriptions **439**. The past values can be based on past results **463** of previous transactions resulting from prior unknown access management events in the IDP. The system **400** can determine differences between the past values and the values of the current access management event **460**, and revise the common purpose of the description **437** based on the differences. For example, a user who frequently provides incorrect passwords when attempting to login from a device with a particular device ID, or a device ID that is associated with their user profile but is not their main device, may have their description revised from being a “potential fraud alert” to being a “failed login.” In another example, the historical record of previous descriptions **439** includes a data-before field, a data-change field, and a data-after field. The system can identify a change in the data-change field, compare that change from the data-after field to the data-before field, and determine differences between the data-after field and the data-before field based on the change.

[0048] FIG. 5 is a flowchart that illustrates a method **500** to contextualize access management events with identity tags and related transaction chains. The method **500** includes monitoring API transactions for an unknown event (step **504**). In some implementations, the API transactions include hooks for streaming output.

[0049] The method **500** includes capturing output from the API transactions in a dynamic record (step **508**). In some implementations, the output includes timestamps, API names, and principals. For example, the output can be captured by streaming the API transactions to the dynamic record using the hooks. The principals can include source information, which in turn can include user ID, session ID, relational ID, device type, device ID, IP address, or client ID.

[0050] The method **500** includes determining related transactions by tagging the transactions with identity tags based on the principals (step **512**). In some implementations, tagging can include parsing the principals for source information and determining profiles that match the source information. The profiles can be included within a profile index hosted by a telecommunications network or by a third-party contracted to the telecommunications network. Each profile of the profiles can correspond to a user or a subscriber of the telecommunications network. For example, users and subscribers can include individuals, families, corporations, companies, or trusts. In some implementations, the profiles include information that identifies the user or subscriber that corresponds to the profiles. Matches can be determined by searching for source information in the profile index. Tags can be determined from the identifying information found in the matching profiles.

[0051] The method **500** includes grouping transactions based on common tags (step **516**). The method **500** includes chaining the related transactions into a list of chained transactions having an order by comparing tags and timestamps and determining proximate timestamps (step **520**). The

method **500** includes determining a prediction of the unknown event by providing the API names in the order of the chained transactions as an input to a model trained to output predicted events based on ordered API names (step **524**). In some implementations, predicting the unknown event includes providing the API names as an input to an intention prediction model trained to output predicted intentions based on API names provided as input. The method **500** can include determining the prediction of the unknown event based on the chained transactions and identity tags.

[0052] The method **500** includes generating a description of the unknown event using a model trained to output descriptions based on predicted events (step **528**). In some implementations, the method **500** includes generating the description for display to a user in a single view. The description can identify a user and a result based on the prediction.

Computer System

[0053] FIG. **6** is a block diagram that illustrates an example of a computer system **600** in which at least some operations described herein can be implemented. As shown, the computer system **600** can include: one or more processors **602**, main memory **606**, non-volatile memory **610**, a network interface device **612**, a video display device **618**, an input/output device **620**, a control device **622** (e.g., keyboard and pointing device), a drive unit **624** that includes a machine-readable (storage) medium **626**, and a signal generation device **630** that are communicatively connected to a bus **616**. The bus **616** represents one or more physical buses and/or point-to-point connections that are connected by appropriate bridges, adapters, or controllers. Various common components (e.g., cache memory) are omitted from FIG. **6** for brevity. Instead, the computer system **600** is intended to illustrate a hardware device on which components illustrated or described relative to the examples of the figures and any other components described in this specification can be implemented.

[0054] The computer system **600** can take any suitable physical form. For example, the computing system **600** can share a similar architecture as that of a server computer, personal computer (PC), tablet computer, mobile telephone, game console, music player, wearable electronic device, network-connected (“smart”) device (e.g., a television or home assistant device), AR/VR systems (e.g., head-mounted display), or any electronic device capable of executing a set of instructions that specify action(s) to be taken by the computing system **600**. In some implementations, the computer system **600** can be an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC), or a distributed system such as a mesh of computer systems, or it can include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **600** can perform operations in real time, in near real time, or in batch mode.

[0055] The network interface device **612** enables the computing system **600** to mediate data in a network **614** with an entity that is external to the computing system **600** through any communication protocol supported by the computing system **600** and the external entity. Examples of the network interface device **612** include a network adapter card, a wireless network interface card, a router, an access point, a wireless router, a switch, a multilayer switch, a protocol converter, a gateway, a bridge, a bridge router, a hub, a digital media receiver, and/or a repeater, as well as all wireless elements noted herein.

[0056] The memory (e.g., main memory **606**, non-volatile memory **610**, machine-readable medium **626**) can be local, remote, or distributed. Although shown as a single medium, the machine-readable medium **626** can include multiple media (e.g., a centralized/distributed database and/or associated caches and servers) that store one or more sets of instructions **628**. The machine-readable medium **626** can include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by the computing system **600**. The machine-readable medium **626** can be non-transitory or comprise a non-transitory device. In this context, a non-transitory storage medium can include a device that is tangible, meaning that the device has a concrete physical form, although the device can change its physical state. Thus, for example, non-transitory refers to a device remaining tangible despite this change in state.

[0057] Although implementations have been described in the context of fully functioning computing devices, the various examples are capable of being distributed as a program product in a variety of forms. Examples of machine-readable storage media, machine-readable media, or computer-readable media include recordable-type media such as volatile and non-volatile memory **610**, removable flash memory, hard disk drives, optical disks, and transmission-type media such as digital and analog communication links.

[0058] In general, the routines executed to implement examples herein can be implemented as part of an operating system or a specific application, component, program, object, module, or sequence of instructions (collectively referred to as “computer programs”). The computer programs typically comprise one or more instructions (e.g., instructions **604**, **608**, **628**) set at various times in various memory and storage devices in computing device(s). When read and executed by the processor **602**, the instruction(s) cause the computing system **600** to perform operations to execute elements involving the various aspects of the disclosure.

Remarks

[0059] The terms “example,” “embodiment,” and “implementation” are used interchangeably. For example, references to “one example” or “an example” in the disclosure can be, but not necessarily are, references to the same implementation; and such references mean at least one of the implementations. The appearances of the phrase “in one example” are not necessarily all referring to the same example, nor are separate or alternative examples mutually exclusive of other examples. A feature, structure, or characteristic described in connection with an example can be included in another example of the disclosure. Moreover, various features are described that can be exhibited by some examples and not by others. Similarly, various requirements are described that can be requirements for some examples but not for other examples.

[0060] The terminology used herein should be interpreted in its broadest reasonable manner, even though it is being used in conjunction with certain specific examples of the invention. The terms used in the disclosure generally have their ordinary meanings in the relevant technical art, within the context of the disclosure, and in the specific context where each term is used. A recital of alternative language or synonyms does not exclude the use of other synonyms. Special significance should not be placed upon whether or not a term is elaborated or discussed herein. The use of highlighting has no influence on the scope and meaning of a term. Further, it will be appreciated that the same thing can be said in more than one way.

[0061] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense—that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” and any variants thereof mean any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import can refer to this application as a whole and not to any particular portions of this application. Where context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number, respectively. The word “or” in reference to a list of two or more items covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list. The term “module” refers broadly to software components, firmware components, and/or hardware components.

[0062] While specific examples of technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative implementations can perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or sub-combinations. Each of these processes or

blocks can be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks can instead be performed or implemented in parallel, or can be performed at different times. Further, any specific numbers noted herein are only examples such that alternative implementations can employ differing values or ranges.

[0063] Details of the disclosed implementations can vary considerably in specific implementations while still being encompassed by the disclosed teachings. As noted above, particular terminology used when describing features or aspects of the invention should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the invention with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the invention to the specific examples disclosed herein, unless the above Detailed Description explicitly defines such terms. Accordingly, the actual scope of the invention encompasses not only the disclosed examples but also all equivalent ways of practicing or implementing the invention under the claims. Some alternative implementations can include additional elements to those implementations described above or include fewer elements.

[0064] Any patents and applications and other references noted above, and any that may be listed in accompanying filing papers, are incorporated herein by reference in their entireties, except for any subject matter disclaimers or disavowals, and except to the extent that the incorporated material is inconsistent with the express disclosure herein, in which case the language in this disclosure controls. Aspects of the invention can be modified to employ the systems, functions, and concepts of the various references described above to provide yet further implementations of the invention.

[0065] To reduce the number of claims, certain implementations are presented below in certain claim forms, but the applicant contemplates various aspects of an invention in other forms. For example, aspects of a claim can be recited in a means-plus-function form or in other forms, such as being embodied in a computer-readable medium. A claim intended to be interpreted as a means-plus-function claim will use the words “means for.” However, the use of the term “for” in any other context is not intended to invoke a similar interpretation. The applicant reserves the right to pursue such additional claim forms either in this application or in a continuing application.

Claims

1. A non-transitory, computer-readable storage medium comprising instructions recorded thereon, wherein the instructions, when executed by at least one data processor of a system, cause the system to: monitor transactions performed by layers of an Identity Provider service (IDP) for an unknown access management event, wherein each layer corresponds to a separate Application Programming Interface (API) with its own distinct functionality in the IDP; capture output from the layers in a dynamic record of the transactions, wherein the output includes timestamps for transactions, names of layers that performed the transactions, and principals responsible for calling the layers; determine related transactions in the dynamic record by tagging the output for the transactions with identity tags based on the principals, and grouping the transactions based on common identity tags; chain the related transactions into a list of chained transactions by comparing identity tags and timestamps and determining an order for the list of chained transactions based on proximate timestamps; determine an access management event based on the names of the layers of the transactions of the list of chained transactions; compose a description of the access management event using a language model trained to output descriptions based on chained transactions as a description input; and generate the description for display to an auditor in a single comprehensive view of the IDP, wherein the single comprehensive view is updated as new output is streamed to the dynamic record of transactions, and wherein the description provides context for the access management event by identifying a user based on the identity tags and a result based on the access management event.

2. The non-transitory, computer-readable storage medium of claim 1, wherein the layers include built-in hooks for streaming output from transactions, and wherein the output from the layers is streamed to the dynamic record of transactions using the built-in hooks.
3. The non-transitory, computer-readable storage medium of claim 1, wherein: the principals include source information, and the source information includes at least one of: user ID; session ID; relational ID; device type; device ID; IP address; or client ID.
4. The non-transitory, computer-readable storage medium of claim 3, wherein tagging the output for each transaction with the identity tags based on the principals further causes the system to: parse the principals for the source information; determine user profiles that match the source information by searching for the source information in a user profile index, wherein the user profiles comprise identifying information; and determine the identity tags from the identifying information.
5. The non-transitory, computer-readable storage medium of claim 1, wherein determining the access management event further causes the system to: provide the names of the layers that performed the transactions as an intention input to an intention prediction model, wherein the intention prediction model has been trained to output predicted intentions based on names of layers provided as intention inputs; determine a predicted intention based on the intention input, wherein the predicted intention includes a common purpose for the transactions; and determine the access management event based on the common purpose and the identity tags.
6. The non-transitory, computer-readable storage medium of claim 5, wherein determining the description of the access management event further causes the system to: organize the list of chained transactions into a hierarchically nested set of feature-value pairs, comprising: features that are descriptive of the unknown access management event; and values associated with the features for each transaction, wherein the hierarchically nested set of feature-value pairs acts as the description input for the language model.
7. The non-transitory, computer-readable storage medium of claim 6, wherein the single comprehensive view comprises a historical record of previous descriptions of access management events performed by the user, wherein the historical record includes a data-before field, a data-change field, and a data-after field, and wherein generating the description for display further causes the system to: identify a change in the data-change field; compare the change in the data-after field against the data-before field; determine differences between the data-after field and the data-before field; and revise the common purpose of the description based on the differences.
8. A system comprising: at least one hardware processor; and at least one non-transitory memory storing instructions, which, when executed by the at least one hardware processor, cause the system to: monitor transactions for an unknown access event, wherein the transactions are performed by separate APIs; capture output from the transactions in a dynamic record, wherein the output includes timestamps, names of APIs that performed the transactions, and principals responsible for calling the APIs; determine related transactions by tagging the output with identity tags based on the principals, and grouping the transactions into the related transactions based on common identity tags; chain the related transactions into a list of chained transactions by comparing the timestamps and determining an order for the list of chained transactions based on proximate timestamps; determine an access event based on the names of the APIs that performed the transactions of the list of chained transactions; compose a description of the access event using a model trained to output descriptions based on chained transactions as input; and generate the description for display to an auditor in a comprehensive view of the transactions, wherein the description identifies a user and a result based on the access event.
9. The system of claim 8, wherein the APIs include built-in hooks for streaming output from transactions, and wherein the output from the APIs is streamed to the dynamic record using the built-in hooks.
10. The system of claim 8, wherein: the principals include source information, and the source

information includes at least one of: user ID, session ID, relational ID, device type, device ID, IP address, or client ID.

11. The system of claim 10, wherein tagging the output for each transaction with identity tags based on the principals further causes the system to: parsing the principals for the source information; determining user profiles that match the source information by searching for the source information in a user profile index, wherein the user profiles comprise identifying information; and determining identity tags from the identifying information.

12. The system of claim 8, wherein predicting a common purpose for the list of chained transactions further causes the system to: providing the names of the APIs that performed the transactions as an input to an intention prediction model, wherein the intention prediction model has been trained to output predicted intentions based on names of layers provided as input; and determine the unknown access event based on the common purpose and the list of chained transactions.

13. The system of claim 12, wherein determining a description of the access event further causes the system to: organizing the list of chained transactions into a hierarchically nested set of feature-value pairs, comprising: features that are descriptive of the unknown access event, and values that correspond with the features for each transaction, wherein the hierarchically nested set of feature-value pairs acts as an input for the model.

14. The system of claim 13, wherein the comprehensive view comprises a historical record of previous descriptions of access management events performed by the user, and wherein generating the description for display further causes the system to: compare the values of the hierarchically nested set of feature-value pairs against past values of the features, wherein the past values are associated with the previous descriptions of the historical record; determine differences between the values and the past values; and revise the common purpose of the description based on the differences.

15. A method comprising: monitor API transactions for an unknown event; capture output from the API transactions in a dynamic record, wherein the output includes timestamps, API names, and principals; determine related transactions by tagging the output with tags based on the principals, and grouping transactions based on common tags; chain the related transactions into a list of chained transactions having an order by comparing tags and timestamps and determining proximate timestamps; determine a prediction of the unknown event by providing the API names in the order of the chained transactions as an input to a model trained to output predicted events based on ordered API names; and generate a description of the unknown event using a model trained to output descriptions based on chained transactions.

16. The method of claim 15, further comprising: generating the description for display to a user in a single view, wherein the description identifies a user and a result based on the prediction.

17. The method of claim 15, wherein the API transactions include hooks for streaming output, and wherein the output from the API transactions is streamed to the dynamic record using the hooks.

18. The method of claim 15, wherein: the principals include source information, and the source information includes at least one of: user ID, session ID, relational ID, device type, device ID, IP address, or client ID.

19. The method of claim 18, wherein tagging the output for each transaction with tags based on the principals further comprises: parsing the principals for the source information; determining profiles that match the source information by searching for the source information in a profile index, wherein the profiles comprise identifying information; and determining tags from the identifying information.

20. The method of claim 15, wherein determining the prediction of the unknown event further comprises: providing the API names as an input to an intention prediction model, wherein the intention prediction model has been trained to output predicted intentions based on API names provided as input; and determine the prediction of the unknown event based on the API names.

